

Sajal Kanti Patra

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Full-Stack Developer with experience building scalable web and cross-platform mobile applications using React Native, Next.js, TypeScript, Node.js, and Supabase. Skilled in REST APIs, real-time systems, AI integration, and Retrieval-Augmented Generation (RAG).

Education

- **RCC Institute of Information Technology (RCCIIT)**

Bachelor in Technology (Computer Science and Engineering); CGPA: 8.46

Kolkata, India

July 2023 – Present
- **Kalaberia Prasanna Kumar High School**

Higher Secondary (stream: science) ; Percentage : 92%

Purba Medinipur, India

Dec 2021 – Dec 2023

Skills Summary

- **Languages:** JavaScript, TypeScript, HTML5, CSS3, Python, C, java.
- **Libraries and Frameworks:** React.js, Next.js, Node.js, Socket.io, Express.js, Chakra UI, Bootstrap, Tailwind CSS, Aceternity UI , Vercel AI sdk , Langchain, Huggingface ,
- **Tools and Platforms:** VS Code, Netlify, Vercel, Git, GitHub, Firebase, Supabase, Postman, Claude Code. ,convex.
- **Database:** MongoDB, MySQL, SQLite , Postgres .

Projects

Next.js, TypeScript, AI tutoring, RAG, real-time streaming, Convex, chat interfaces

Class-Pilot

- Built an AI-powered learning platform that supports course-based tutoring, interactive chat, and personalized study assistance.
- Implemented retrieval-augmented generation to provide context-aware answers from course material and learning resources.
- Added real-time streaming workflows for responsive tutor interactions and a smoother student experience.
- Designed modular course and tutor pages to support scalable academic workflows and content delivery.

MessMate

TypeScript, React Native, Expo, Supabase, NativeWind, Reanimated.

- Full-Stack Mobile Expense Tracking App with Real-time Sync
- Built a group expense management app for roommates and travel groups to track shared costs, split expenses flexibly, and settle debts through an optimized payment algorithm.
- Implemented a greedy settlement engine that minimizes transactions needed to clear all group debts, plus three split modes (equal/custom/weighted) with multi-payer support.
- Designed real-time synchronization via Supabase subscriptions for instant cross-device updates, with push notifications for expense alerts and settlement reminders
- Built analytics dashboard with spending breakdowns, member comparisons, trend charts, and Excel export functionality —wrapped in a modern glassmorphic dark UI.

CivicApp

TypeScript, React Native, Expo, Next.js, Supabase, Clerk, Tailwind CSS, AWS S3, Cloudinary

- Developed a cross-platform civic issue reporting system with a mobile app for citizens and a real-time admin dashboard for municipalities.
- Built offline-first reporting with GPS location tagging, complaint tracking, and push notifications for status updates.
- Implemented duplicate issue detection using perceptual hashing and geo-proximity analysis to reduce redundant reports.
- Created an auto-priority scoring system to classify issue

Expense Tracker AI

Next.js 16, TypeScript, React 19, Prisma, PostgreSQL, Clerk, Recharts, TailwindCSS 4

- Full-Stack Personal Finance Management App with Smart Analytics
- Built an intelligent expense tracking platform with secure authentication (Clerk), server-side rendering, and optimized database queries using Prisma ORM with PostgreSQL on Neon.
- Implemented 8 smart features: real-time budget tracking with alerts (green/yellow/red status), 6-month spending trends, category-based analytics, CSV export, and advanced search/filtering with multi-criteria support.
- Designed financial insights engine with month-over-month comparison, average daily spending calculations, and AI-powered tips—featuring interactive pie/line charts via Recharts and dark mode support.
- Architected with Next.js App Router using server actions for optimized performance, React Server Components for reduced bundle size, and responsive UI with glassmorphic design elements